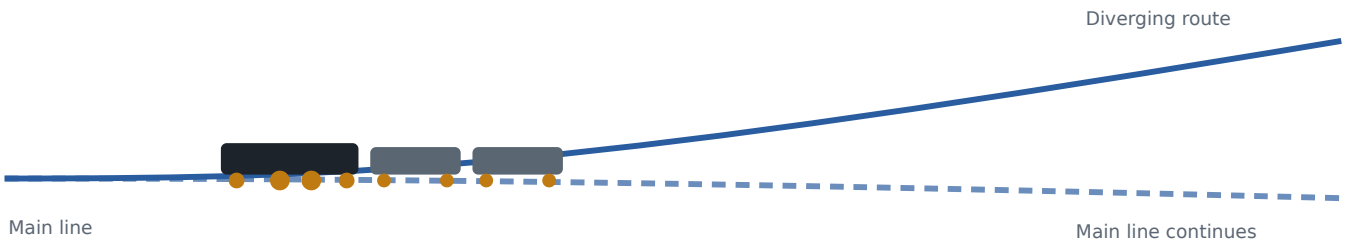
BLENDER 3D TRAIN ANIMATION TOOLKIT

Rail Animator Track It Framework

A concept guide and completion plan: architecture, track and switching systems, vehicles, rigs, consists, the asset workflow, and a worked 2-4-2 steam locomotive with a two-bogie tender.



Subject	“Track It” Blender add-on (`Track it compiled 1.py` , bl_info 1.32, 3,143 lines), plus design sessions 1–6
Sources reviewed	Handoff v8 · the latest add-on source · sessions on track streamlining, curve dragging, nearest-curve selection, switching mechanism, vehicle architecture
Document type	Concept and planning only. Nothing in here has been built or tested in Blender unless it is marked **Built**
Date	26 September 2026

PART 0

How to use this guide

The guide is split into standalone Parts, one per major topic, and each Part is also delivered as its own PDF. Read Parts 1–3 first. After that, jump to whatever you're working on.

The Parts

Part	Topic	What you get from it
0	Introduction	This page, how to read the callouts, glossary.
1	Where the project stands	History from the old Clamp To / shrinkwrap rig to Follow Path. A code audit of the current add-on, feature status, and every issue found.
2	**Curves explained (visual)**	How a Blender curve is built, how Follow Path turns it into motion, what offset_factor really measures, and how Track It adds attach, drag, switching and pivots. Start here if curves feel like a black box.
3	Framework architecture	The four-layer model (Track → Mechanism → Rig → Mesh), the target data model, the per-frame evaluation pipeline, and the proposed package layout.
4	Track system	One-spline rule, arc-table caching, the Track Network graph, overlap zones, joints, banking, and the Track Builder and Track Doctor tools.
5	Switching and route history	How influence switching works now, why two history systems are one too many, and the proposed **Turnout-state + Route Plan** model.
6	Vehicles and pivots	Pivot roles, root placement, body orientation, pivots crossing curves, binding a rig to pivots, and the Vehicle Definition.
7	Motion, drive and wheels	The distance-driven motion model, drive controls, wheel modes (Auto / Direct / Slip), secondary motion (sway, hunting, suspension), and render-safe evaluation.
8	Consist system	Cars following a lead, coupling and uncoupling, driving from any car, and reversing into a siding.
9	Asset workflow	Step by step: turning a plain unrigged model into a Track It vehicle asset in the library.
10	Concept study: 2-4-2 + tender	Rigging an unrigged 2-4-2 steam locomotive with a two-bogie tender and bringing it into the framework.

Part	Topic	What you get from it
11	Roadmap	Phased completion plan, prioritised backlog, testing strategy, and the decisions still open.

Reading the callouts

CURRENT STATE
What the add-on does *today*, taken from the source file and handoff v8.

PROPOSAL
A new idea or improvement from this review. Not built yet.

ISSUE / RISK
A bug or fragile spot found in the code, or a known Blender pitfall (the modal-transform crash, for example).

DECISION NEEDED
A design choice that is yours to make before building.

RIG LAYER
Violet marks the artist-facing control rig: armatures, bones, and the drivers the animator touches.

Diagrams use one colour key throughout: **blue = track / curves / mechanism**, **amber = pivots**, **violet = rig**, **green = proposed / new**, **red = problem**.

Glossary

Term	Meaning in this project
Track / curve / spline	A Blender Curve object holding exactly one spline that a vehicle can ride. Follow Path only ever reads `splines[0]`.
offset_factor	The Follow Path value (0–1, with `use_fixed_location`) that places an object along a curve. It equals *distance ÷ curve length*: it is arc-length based.
Arc table	The add-on's 5,000-sample lookup of cumulative length vs world position, used to convert a world position into an offset_factor.
Held constraint	A Follow Path constraint the object keeps for a curve it has been near. Influence 1 = driving, 0 = inert but kept (its keyframes stay intact).
Overlap zone	A stretch where two curves run on top of each other, so a vehicle can change curve without a visible pop.

Term	Meaning in this project
Divergence point	Where two overlapping curves separate. After it, switching between them is physically impossible.
Joint	End-to-end connection between two curves with no choice (one way on).
Turnout / junction	A real fork where a decision is needed. Proposed as its own object with a keyframeable state (Part 4).
Vehicle root	The object the add-on actually drives along the track. The rig and mesh ride on it and are never bolted to it permanently.
Pivot	A point of the vehicle that tracks the rail on its own, such as a bogie, pony truck or trailing truck. Currently an auto-managed Empty with its own Follow Path.
Consist	A train: a lead vehicle plus coupled cars, each a vehicle with its own pivots.
Route distance / odometer	The running distance (metres) a vehicle has travelled along its actual route, across curve changes. The basis of the route-history log.
Live switching	Changing curves while dragging, with no keyframe written. Only at recorded forks or at a single unambiguous continuation.
Drag Slider	The spring-centred N-panel slider that nudges offset_factor. It is the primary live control; Mouse Drag is deprecated.
2-4-2	Whyte notation: 2 leading wheels (1 axle), 4 driving wheels (2 axles), 2 trailing wheels (1 axle).

THE RULE EVERY PART RESPECTS

Never create or write constraints on an object from a background handler while a 3D transform (`TRANSFORM_OT_translate`) is modal. This caused three identical crashes inside `DEG_id_tag_update → recalcData_objects → transform_modal`. It is why Mouse Drag is deprecated and why live control goes through N-panel widgets.